Downloaded samplesgshA1.txt file

1.) Check quality with fastqc, but using falco (Galaxy Version 1.3.0+galaxy0) b/c faster
   - Default parameters
   - Run with transcriptome reads
      - Forward data
         - Fails: per tile sequence quality; Sequence duplication levels; Adapter Content
         - Warnings: Per base sequence content; Per sequence GC content; Overrepresented sequences
      - Reverse Data
         - Fails: Per tile sequence quality; adapter content
         - Warnings: Per base sequence content; Per sequence GC content; Sequence duplication levels; Overrepresented sequences

2.) Trim with fastp (Galaxy Version 1.3.3+galaxy0)
   - Set to paired collection
   - Gets rid of stuff that are fails for the quality control
      - Observe the results on HTML, those under filtering reports

3.) HISAT2 (Galaxy Version 2.2.2+galaxy0) - aligns transcriptome reads to the genome that is provided by Dr. Van Laar
   - Change in parameters:
- Source for reference genome: genome from the history
- Single or paired library? Set to paired-end dataset collection

Made an alignment file called BAM

Now, what actual transcripts do we have?

Step4.) StringTie ((Galaxy Version 2.2.3+galaxy0) - assembles short reads into actual transcripts, predicts full length transcripts
- Make sure input option parameter is set to Short Reads

Created 6 files of transcripts
- Change parameter to reference genome to history
   - Burkholderia.gff

Step 4.) StringTieMerge (Galaxy Version 2.2.3+galaxy0) - instead of 6 files, puts them all together
- Use dataset collection and use the StringTie file
- Use reference as burkholderia.gff

- step 5.) featureCounts (Galaxy Version 2.1.1+galaxy0)
   - Set alignment file to dataset collection and use HISAT file
   - Set gene annotation file as GFF/GTF from history and use latest stringtiemerge
   - Rewrite GFF gene identifier as transcript_id

- Does the input have read pairs? Set to Yes, paired-end and count them as 1 single fragment
- Next step: Deseq2 (Galaxy Version 2.11.40.8+galaxy2) - function? Takes all 6 conditions and sees if it is more represented in one condition or the other, gives statistical significance
    - Make sure to rename to Counts under featurecounts for easier setup
- Factor: set to Gene_Type
- 2 factor levels. Factor level 1 set as WildType; Factor level 2 set as Mutant
    - Rename the counts as wildtype or mutants
    - Tells if gene was up or down regulated
        - Under Advanced options:
    - Use beta priors? Set to yes
    - Under Output options: generate plots and output normalised counts
    - Click “include hidden” within history and locate the featurecount and find the Counts, rename as plate/BrothReplicate#
    - Put these in using the multiple dataset options, search with the ellipses
- From the result file you get a p-value, can determine null hypothesis
    - For the gene, there is no change in gene expression between wildtype and mutant: null hypothesis
    - If there are changes between the two, you can infer significant difference, but need this p-value to make a statistical backing for your reasoning
    - Less than 0.05, it is statistically significant

downloaded both the plot and the result file

Step 6.) **filter data on any column using simple expressions**

- Use on DEseq2 file
- With following condition - c7<0.05
    - Column 7 is the p-adj, run it on this one instead of the raw p-value column
- Run again with the following condition - abs(c3)>1
    - Column 3 is the log2(FC), run on this
    - Only do this after the previous filtering is finished
- From the plots we download here, we learn about whether expression change between the two growth media/conditions is statistically significant enough to reject the null hypothesis

Step 7.) filter on dataset with following condition:

- abs(c3)>1
    - Cut it down to 626 lines from 1010 lines

Step 8.) were gonna add a header now through a new Excel file

- Put the following in: GeneID Base mean log2(FC) StdErr Wald-Stats P-value P-adj

Step 9.) concatonate tool

- Insert the most recent filter on dataset file

Now our data file has proper headers!

- Rename the file to Differential Gene Expression of Mutant and Wildtype
- Make sure to save this as a csv file!

- Go to edit attributes > datatypes > target datatype: tabular (using ‘Convert CSV to Tabular’)

Step 10.) volcano plot (Galaxy Version 4.0.3+galaxy0) - represents p-values
- Change the respective columns into the following: 7, 6, 3, 1
- Downloaded this

Step 11.) opened the tabular broth vs plate file as an excel file
- Sort the entire document from smallest to largest in log2(FC), kept the top 10 and bottom 10, delete everything in between, then save as a .csv file (separately so that you can still have the initial file)
    - CTRL a > data > sort > input conditions log2(FC) and smallest to largest
    - Make sure you delete all the cells that arent included in the list

Step 12.) upload that new.csv file to galaxy
- Edit attribute on galaxy > datatypes > make sure New Type is set as “tabular”

Step 13.) join 2 datasets side by side on a specified field (Galaxy Version 2.1.3)
- Join .csv file with normalized counts dataset
- Parameter change: Keep Header Lines - Yes

Step 14.) cut columns from a table Galaxy Version 1.0.2)
- parameter : cut c8-c14
    - *may need to set as c9-c14 if there is a blank column, check by clicking on the file in galaxy and scrolling to the right to see if there is an additional column*
    - *didnt need to make adjustments, c8-c14 was fine*

Step 15.) heatmap2 - (Galaxy Version 3.3.0+galaxy0)
- Change parameters:
    - Data transformation - Log2(value+1) transform my data
    - Blue (high level expression) Wildtype has expression in the top half of the genes, and the same goes for the mutants
    - Brown (low level of expression) Wildtype has lower levels of expression in the bottom half of the genes, and the same goes for the mutants
    - Will download this file as well
        - For the mutant type and the wildtype, they have high expression for this gene: *BTH_RS25860*
            - Deals with the distribution of pilus subunits, may affect the ability of DNA transfer, adhesion to host cells, and perhaps biofilm formation
        - Low expression for this gene: *BTH_RS29935*
            - Sugar metabolism and stress regulator, perhaps affects survivability of the organism and what how sugars are processed (maybe what kind of sugars they are able to effectively metabolize)

Step 16.) filter data on any column using simple expressions (Galaxy Version 1.1.1)

- Filter - burkholderia.gff
- Wanna keep all under CDS under column 3
    - With following condition: c3=='CDS'

Now we wanna keep column 9

Step 17.) Cut column from a table (Galaxy Version 1.0.2)
- Retain only column 9
- Parameters: cut columns - c9

Cant read these intuitively so…

Step 17.) convert delimiters to TABGalaxy Version 1.0.1)
- Parameters:
    - Convert all - semicolons

We just want the gene name on column 2

Step 18.) replace text in a specific column (Galaxy Version 9.5+galaxy3)
- Parameters:
    - In column 2
    - Find pattern - Parent=
- Can preview under visualize for confirmation
    - Yup, confirmed it was taken out

Now we're interested in column 8 because it has the gene function
We want the gene name column 2 and the gene products column 8

Step 19.) cut columns from a table (Galaxy Version 1.0.2)
- Parameters:
    - Cut columns: c2,c8

Step 20.) join two datasets side by side on a specified field (Galaxy Version 2.1.3)
- Join cut on dataset with latest concatenate.csv file (the one you set as tabular)
- Keep header lines - yes